

Date	31 January 2026
Team Id	LTVIP2026TMIDS86939
Project Name	Visualizing housing market trends
Maximum Marks	4 marks

BRAINSTORMING – IDEA GENERATION & PRIORITIZATION

1. Idea Generation

- Then below this paste cheyyi:
- Analyze average sale price of houses
- Compare sale price based on number of bedrooms
- Study relationship between flat area and sale price
- Identify high price and low price segments
- Develop interactive Tableau dashboards

2. Idea Prioritization

After evaluating impact and feasibility, the following ideas were selected:

- Average Sale Price by Bedrooms
- Sale Price vs Flat Area
- Combined Dashboard Visualization

These ideas were selected because they are easy to implement in Tableau and provide high analytical value.

Define Problem Statement

Problem Statement

Understanding housing market trends is challenging due to large volumes of data and multiple influencing factors. Buyers and analysts often struggle to identify how features such as number of bedrooms and flat area affect sale prices.

There is a need for a clear and interactive data visualization system that helps users analyze housing trends efficiently. This project aims to use Tableau to create meaningful dashboards that provide insights into housing market patterns.

Empathy Map Canvas

Target User:

Home Buyer / Real Estate Analyst

SAYS:

- I want to know how house prices vary.
- Which house gives better value for money?
- Does area affect sale price?

THINKS:

- Is this property worth its price?
- Bigger house means higher price aa?
- Which option is the best investment?

DOES:

- Compares house prices online
- Check number of bedrooms and area
- Analyzes market trends before buying

FEELS:

- Confused due to large amount of data
- Worried about making wrong decision

- Wants clear and simple visual insights

SOLUTION REQUIREMENTS

1. Introduction

This project focuses on analyzing housing market data to understand the relationship between sale price and property features using Tableau.

2. Existing System

Currently, housing data is available in raw format such as Excel sheets or databases. It is difficult for users to interpret trends without proper visualization.

3. Proposed System

The proposed system uses Tableau to create interactive dashboards that visualize housing trends and provide insights based on number of bedrooms and flat area.

4. Functional Requirements

- Import dataset into Tableau
- Create charts and graphs
- Develop interactive dashboard
- Present insights using story feature

5. Non-Functional Requirements

- User-friendly interface
- Accurate data representation
- Fast performance

TECHNOLOGY STACK

Tools Used

- Tableau – Data Visualization
- Microsoft Excel – Data Preparation
- GitHub – Version Control and Hosting

MS Word – Documentation

Data Flow Diagram

Explanation:

- Housing dataset collected
- Data cleaned in Excel
- Dataset imported into Tableau
- Charts and dashboard created
- Insights delivered to users

Simple Diagram:

1.Data Source

2.Data Cleaning

3.Tableau

4.Dashboard

5. User

User Stories

- As a home buyer, I want to see how prices change based on area so that I can make better decisions.
- As an analyst, I want to compare price trends based on bedrooms.
- As a user, I want an interactive dashboard for easy understanding.

PROBLEM SOLUTION FIT

Problem

Home buyers and analysts find it difficult to understand housing market trends due to large and unstructured data. It is hard to analyze how factors like number of bedrooms and flat area affect sale prices.

Target Users

- Home Buyers
- Real Estate Analysts
- Real Estate Agencies
- Proposed Solution

Develop an interactive Tableau dashboard that visualizes housing sale prices based on features like bedrooms and flat area.

Value Proposition

The dashboard provides clear insights, easy comparison of house prices, and better decision-making support through visual representation.

PROPOSED SOLUTION

This project proposes a data visualization solution using Tableau to analyze housing market data. The system imports dataset into Tableau, performs analysis, and generates interactive dashboards.

The dashboard includes:

- Average Sale Price by Bedrooms
- Sale Price vs Flat Area
- Combined Dashboard
- Story presentation explaining insights

This solution helps users easily interpret housing trends and make informed decisions.

Solution Architecture

Architecture Explanation

- Housing dataset collected
- Data cleaned using Excel
- Data imported into Tableau
- Analysis performed using charts and graphs
- Dashboard and story created
- Insights delivered to users

Simple Architecture Flow

- 1.Data Source
- 2.Excel Processing
- 3.Tableau
- 4.Dashboard
- 5.End User

Planning Logic

The project was planned in a structured and systematic manner to ensure successful completion.

The planning process included:

- 1.Understanding project objectives
- 2.Collecting and preparing dataset
- 3.Performing data analysis in Tableau
- 4.Creating visualizations and dashboards
- 5.Interpreting insights

6.Preparing documentation

7.Final testing and submission

This logical sequence helped in completing the project efficiently within the given timeline.

Project Planning

Phase 1 – Ideation

- Identified project topic
- Defined problem statement
- Generated and prioritized ideas

Phase 2 – Requirement Analysis

- Defined functional and non-functional requirements
- Identified technology stack
- Created data flow explanation

Phase 3 – Design

- Designed solution architecture
- Prepared problem-solution fit
- Planned dashboard structure

Phase 4 – Implementation

- Imported dataset into Tableau
- Created charts and dashboards
- Developed story presentation

Phase 5 – Testing & Documentation

- Verified dashboard outputs
- Prepared final documentation
- Uploaded files to GitHub

USER ACCEPTANCE TESTING (UAT)

User Acceptance Testing was conducted to verify that the dashboard meets user requirements.

Test Results:

- Users were able to understand housing trends clearly.
- Visualizations were easy to interpret.
- Interactive dashboard helped in comparing sale prices effectively.
- The system fulfilled the project objectives successfully.

The users confirmed that the dashboard provides meaningful insights and supports decision-making.